Expressions & Equations Error Analysis

TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Distributing

ORIGINAL PROBLEM

$3(x + 4) = 21$

INCORRECT WORKED SOLUTION

$3(x + 4) = 21$

$3x + 4 = 21$

$3x = 17$

$x = \frac{17}{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

The 3 was multiplied by x but not by 4. The distributive property multiplies every term inside the parentheses.

B. SHOW CORRECT WORK (1 POINT)

$3(x + 4) = 21$

$3x + 12 = 21$

$3x = 9$

$x = 3$

2. Combining unlike terms

ORIGINAL PROBLEM

$5x + 3 + 2x = 24$

INCORRECT WORKED SOLUTION

$5x + 3 + 2x = 24$

$8x + 2x = 24$

$10x = 24$

$x = \frac{12}{5}$

A. EXPLAIN THE MISTAKE (1 POINT)

5x and 3 are unlike terms, so they cannot be added together. Only the x-terms (5x and 2x) combine.

B. SHOW CORRECT WORK (1 POINT)

$5x + 3 + 2x = 24$

$7x + 3 = 24$

$7x = 21$

$x = 3$

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3. Undoing subtraction

ORIGINAL PROBLEM

$2x - 7 = 15$

INCORRECT WORKED SOLUTION

$2x - 7 = 15$

$2x = 15 - 7$

$2x = 8$

$x = 4$

A. EXPLAIN THE MISTAKE (1 POINT)

To undo "− 7" you must ADD 7 to both sides. Subtracting 7 from the right side gives the wrong equation.

B. SHOW CORRECT WORK (1 POINT)

$2x - 7 = 15$

$2x = 22$

$x = 11$

4. Dividing every term

ORIGINAL PROBLEM

$6x + 9 = 33$

INCORRECT WORKED SOLUTION

$6x + 9 = 33$

$\frac{6x + 9}{3} = \frac{33}{3}$

$2x + 9 = 11$

$2x = 2$

$x = 1$

A. EXPLAIN THE MISTAKE (1 POINT)

Dividing both sides by 3 means dividing EVERY term by 3. The 9 should have become 3.

B. SHOW CORRECT WORK (1 POINT)

$6x + 9 = 33$

$\frac{6x + 9}{3} = \frac{33}{3}$

$2x + 3 = 11$

$2x = 8$

$x = 4$

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5. Variables on both sides

ORIGINAL PROBLEM

$7x - 4 = 3x + 12$

INCORRECT WORKED SOLUTION

$7x - 4 = 3x + 12$

$7x + 3x - 4 = 12$

$10x - 4 = 12$

$10x = 16$

$x = \frac{8}{5}$

A. EXPLAIN THE MISTAKE (1 POINT)

Whatever you do to one side you must do to the other. To remove 3x from the right side, SUBTRACT 3x from both sides ($7x - 3x = 4x$).

B. SHOW CORRECT WORK (1 POINT)

$7x - 4 = 3x + 12$

$4x - 4 = 12$

$4x = 16$

$x = 4$

6. Distributing a negative

ORIGINAL PROBLEM

$5 - (x - 3) = 10$

INCORRECT WORKED SOLUTION

$5 - (x - 3) = 10$

$5 - x - 3 = 10$

$2 - x = 10$

$-x = 8$

$x = -8$

A. EXPLAIN THE MISTAKE (1 POINT)

The subtraction sign changes BOTH terms in the parentheses: $-(x - 3) = -x + 3$, not $-x - 3$.

B. SHOW CORRECT WORK (1 POINT)

$5 - (x - 3) = 10$

$5 - x + 3 = 10$

$8 - x = 10$

$-x = 2$

$x = -2$

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7. Clearing a fraction

ORIGINAL PROBLEM

$\frac{x}{4} + 3 = 8$

INCORRECT WORKED SOLUTION

$\frac{x}{4} + 3 = 8$

$x + 3 = 32$

$x = 29$

A. EXPLAIN THE MISTAKE (1 POINT)

When both sides are multiplied by 4, every term gets multiplied, including the 3. It should become 12.

B. SHOW CORRECT WORK (1 POINT)

$\frac{x}{4} + 3 = 8$

$\frac{x}{4} = 5$

$x = 20$

8. Order of operations

ORIGINAL PROBLEM

Evaluate $2x^2 - 3$ for $x = -4$

INCORRECT WORKED SOLUTION

$2x^2 - 3$

$= (2 \cdot (-4))^2 - 3$

$= (-8)^2 - 3$

$= 64 - 3 = 61$

A. EXPLAIN THE MISTAKE (1 POINT)

The exponent 2 applies only to x, not to 2x. Square −4 first, then multiply by 2.

B. SHOW CORRECT WORK (1 POINT)

$2x^2 - 3$

$= 2(-4)^2 - 3$

$= 2(16) - 3$

$= 32 - 3 = 29$

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9. Identity equation

ORIGINAL PROBLEM

$2(x + 3) = 2x + 6$

INCORRECT WORKED SOLUTION

$2(x + 3) = 2x + 6$

$2x + 6 = 2x + 6$

$6 = 6$

$x = 6$

A. EXPLAIN THE MISTAKE (1 POINT)

6 = 6 is always true, which means EVERY value of x works. The 6 is not the value of x. There are infinitely many solutions.

B. SHOW CORRECT WORK (1 POINT)

$2(x + 3) = 2x + 6$

$2x + 6 = 2x + 6$

$6 = 6$ (always true)

All real numbers (infinitely many solutions)

10. Solving a formula

ORIGINAL PROBLEM

Solve $P = 2l + 2w$ for w.

INCORRECT WORKED SOLUTION

$P = 2l + 2w$

$P - 2l = 2w$

$w = \frac{P}{2} - 2l$

A. EXPLAIN THE MISTAKE (1 POINT)

Dividing by 2 must be applied to the WHOLE left side ($P - 2l$), not just to P.

B. SHOW CORRECT WORK (1 POINT)

$P = 2l + 2w$

$P - 2l = 2w$

$w = \frac{P - 2l}{2}$

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